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1 November 1987
SUPERSEDING
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MILITARY SPECIFICATION

AEROSOL GENERATOR, NON-THERMAL, INSECTICIDE, ULTRA LOW VOLUME

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a gasoline-engine-driven nonthermal aerosol generator for applying insecticides in ultra low volume (ULV) operations.

1.2 Classification.

Type I. Designed for installation use.

Type II. Designed for field (off-road) use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Biomedical Research and Development Laboratory, ATTN: SGRD-UBE-D, Fort Detrick, Frederick, MD 21701-5010, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3740

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ENCL

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-514	-	Plates, Identification, Instruction and Marking Blank.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-L-2104	-	Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-F-3541	-	Fittings, Lubrication.
MIL-G-3859	-	Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-G-10924	-	Grease, Automotive and Artillery.
MIL-S-12651	-	Sprayers, Packaging of.
MIL-G-23827	-	Grease, Aircraft and Instrument, Gear, and Actuator Screw.
MIL-L-46167	-	Lubricating Oil, Internal Combustion Engine, Arctic.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-1410	-	Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1474	-	Noise Limits for Army Materiel.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

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2.1.2 (cont'd)

DEPARTMENT OF LABOR

29 CFR 1900-1999	-	Occupational Safety and Health Administration, Department of Labor.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Reg. No. 8329-20	-	Dursban® 1.5 ULV Mosquitocide (chlorpyrifos).
EPA Reg. No. 241-208-AA	-	Cythion® (technical grade malathion).

(Copies of specifications, publications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J429	-	Mechanical and Material Requirements for Externally Threaded Fasteners.
J537	-	Storage Batteries.
J1174	-	Operator Ear Sound Level, Measurement Procedure for Small Engine Powered Equipment

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1	-	Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

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2.2 (cont'd)

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The aerosol generators furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.2.1.).

3.2 Description. The aerosol generator shall consist of a metering system which delivers insecticides at flow rates of 1 1/3 to 18 fluid ounces per minute, an engine fueled by gasoline and equipped with an electric starter-generator, discharge nozzle(s) with 180 degree horizontal and vertical adjustments, a minimum 10 gallon nonpressurized, noncorrosive insecticide tank which is easily removable with adequate provisions for drainage, a fuel tank with at least a 2 1/2 gallon capacity, and other accessories necessary for a safe and satisfactory operational unit. The unit shall be designed for remote control and one-man operation.

3.3 Design. The aerosol generator shall be designed in accordance with MIL-STD-1472 for the purpose of dispensing all EPA registered concentrated insecticides for ULV ground applications. The machine shall be capable of dispensing all such insecticides in accordance with EPA label requirements. The aerosol generator design shall meet Occupational Safety and Health Administration (OSHA) requirements for the protection of operating personnel, including 29 CFR 1910.219, Mechanical Power Transmission Apparatus. Concentrated insecticides shall be taken from a tank through the system (which includes an 80-mesh, or finer, stainless steel screen filter) by means of an adjustable constant-volume pump. The filter shall be placed near the insecticide tank so that the formulation is filtered as it is drawn from the tank. A control panel shall be provided whether on the machine or on the remote control box equipped so that a quick readout of the formulation flow rates may be adequately monitored. A liquid filled pressure gauge shall be required to monitor system air pressure essential to the maintenance droplet size. No insecticide lines shall be placed in the cab of the vehicle which carries the aerosol generator. During operation, the insecticide shall flow

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3.3 (cont'd)

to the dispensing head(s) where it will be atomized and dispersed into the atmosphere. When specified (see 6.2), gauges shall be provided for insecticide and fuel tanks.

3.3.1 Size and volume. Type I units shall have a 14-18 HP engine with a blower size rating of 250-350 cubic feet per minute (CFM). Type II units shall be capable of being loaded/unloaded by four people on the appropriate vehicle and shall have a 8-13 HP engine with a blower size rating of 125-249 CFM (see 6.2).

3.4 Material. All insecticide conveying parts shall be made of stainless steel, nylon, polyethylene, viton, polytetrafluroethylene, polypropylene acetal, or other suitable corrosion resistant materials. Seals and O-rings within the insecticide system shall not be made of neoprene. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product (see 6.7).

3.5 Identical items. All aerosol generator units furnished under a specific contract shall be physically and mechanically identical, in that all parts, components, accessories, and subassemblies shall be interchangeable on all aerosol generator units. No deviation shall be acceptable without prior written approval from the contracting officer.

3.6 Performance. The aerosol generator shall operate satisfactorily at designed engine speed, shall not exceed 80 percent of maximum rated horsepower at that speed, and shall operate continuously, except for refueling, for the hours specified in section 4.

3.7 Droplet size. All EPA registered insecticides for ULV aerosol use shall be atomized according to 4.5.2.5.

3.8 Noise limits.

3.8.1 Personnel noise. The noise level at the operator's station when the aerosol generator is operated shall not exceed 85 decibels (dB(A)) when tested in accordance with 4.5.2.7.1.

3.8.1.1 Precautions (noise). Hearing caution signs shall be posted on equipment if the sound level exceeds 85 dB(A) at the operator's station when the aerosol generator is operated. Noise level of 85 dB(A) may only be exceeded when it is beyond the present state-of-the-art. Caution signs, if required, shall state "CAUTION, HEARING PROTECTION REQUIRED WITHIN (SPECIFY) FEET". Specify the maximum distance in feet that 85 dB(A) are measured from the equipment. Plates shall be in accordance with MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material.

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3.8.2 Exterior noise. The aerosol generators noise level shall comply with MIL-STD-1474 when tested in accordance with 4.5.2.7.2.

3.9 Safety. All shafts or rotating parts shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment.

3.9.1 Additional safety features. Adequate shielding may be used to attain the same results specified in 3.9. Exhausts or discharges from the equipment shall be directed so that they do not endanger personnel. Electrical equipment shall be effectively guarded and bonded to protect all persons and objects from electrical shock hazards.

3.10 Roadability. When properly loaded on a carrier vehicle, the aerosol generator shall be operated while the carrier vehicle is moving on unimproved roads and cross-country trails. During or following any driving operation, there shall be no damage or displacement of any component, accessory, or part in or on the aerosol generator (see 4.5.2.3).

3.11 Reliability. The aerosol generator shall be tested for compliance with reliability criteria as specified in 4.5.2.8.

3.12 Maintenance ratio. The aerosol generator shall have a maintenance ratio of not more than 0.05. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing (see 4.5.2.9).

3.13 Engine. For the purpose of selecting the engine, the aerosol generator shall be classified as a class I end item in accordance with MIL-STD-1410. The gasoline engine including all systems components and accessories shall conform to MIL-STD-1410, class I, except as otherwise specified herein. The engine shall be capable of operating under all conditions specified herein when using fuels and lubricants specified herein and in MIL-STD-1410. Additionally, engine horsepower shall not exceed 80 percent of maximum rated horsepower during normal operations or during either the performance or spray analysis test.

3.13.1 Electromagnetic interference characteristics. When specified (see 6.2), the electromagnetic interference emission characteristics of the aerosol generator shall conform to MIL-STD-461, class II (see 4.5.2.6).

3.14 Battery. A battery conforming to SAE J537 with appropriate universal mounting battery box shall be furnished. The top of the battery shall be shielded to avoid accidental shorting of the battery.

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3.15 Blower or Air Pump. The blower or air pump shall be capable of providing adequate air pressure and volume for atomization of all EPA ULV registered pesticides when tested in accordance with 4.5.2.5. The engine shall operate at a speed which does not exceed maximum rated revolutions per minute (rpm) and at a power level which does not exceed 80 percent of maximum rated horsepower.

3.16 Fuel tank. The fuel tank shall have a minimum 2 1/2 gallon capacity and shall enable the aerosol generator to operate for at least 3 continuous hours without refueling. The cap of the fuel tank shall be designed to prevent fuel seepage when the tank is filled (to the bottom of the filler neck) and the engine is running at design speed.

3.17 Remote control unit. The aerosol generator shall be equipped with a mechanical or electronic remote control unit. The cable between the machine and remote control panel shall be a minimum of 16 feet in length and shall be encased in flexible conduit. All controls shall be readily accessible and within easy reach and sight of the operator. The panel shall be equipped with a night light that illuminates all instrumentation on the panel.

3.17.1 Instrumentation. The control system shall be equipped with a mechanical or solid-state electronic flow control having an illuminated digital or analog readout, flow indicator, and flow ON-OFF switch. Instrumentation should be protected against reverse polarity by suitable electronic devices. The machine shall also be equipped with an air pressure gauge and ON-OFF switch (provision for instantaneous engine shutdown). Insecticide shall not flow through lines in proximity to the operator/vehicle driver. When specified (see 6.2), an elapsed time indicator to read operational time of the unit shall be furnished as part of the instrumentation.

3.17.1.1 Calibration instructions. The manufacturer shall provide specific guidance on all machine settings such as pressure and engine rpm.

3.17.1.2 Durability. Instrumentation shall be protected from vibration, shock, and effects of insecticide, and satisfy all provisions of 4.5.2.

3.18 Flushing system. The aerosol generator shall incorporate a flushing system which includes a solvent tank that is connected in parallel with the insecticide tank. The system shall be equipped with a valve so that flushing is accomplished by manually changing the valve position from insecticide to solvent. The system shall be designed to be in-line so that the system can be changed from dispensing insecticides to flushing by use of a selector valve.

3.19 Lubrication.

3.19.1 Lubricants. When applicable, the aerosol generator shall operate as specified herein when serviced with the following military lubricants:

Engine: MIL-L-2104 or MIL-L 46167

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3.19.1 (cont'd)

Grease Lubrication: MIL-G-10924 or MIL-G-23827

3.19.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.20 Treatment and painting. The portions of the aerosol generator normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A color, as specified (see 6.2). Paint shall be resistant to damage by insecticide chemicals.

3.21 Caution markings. In addition to caution markings, when hearing protection is required, each unit shall be prominently marked as follows: PERSONAL PROTECTIVE CLOTHING AND EQUIPMENT MUST BE WORN WHILE ULV MACHINE IS OPERATING. Markings shall be solvent-resistant and wear-resistant so that they remain legible during the expected use-life of the unit.

3.22 Identification marking. The aerosol generator shall be identified in accordance with MIL-STD-130. The marking shall be applied to the aerosol generator on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material, 0.020 inch thick.

3.22.1 Additional marking. The gross weight of each aerosol generator shall be stenciled on each unit. Applicable hazard precautions and applicable references to operating instruction documents shall be applied by stenciling or labeling (see 3.8 and 3.21). Stenciling shall conform to MIL-T-704.

3.23 Workmanship. The aerosol generator shall be clean and free from sand, dirt, fins, pits, scale, rust, burrs, sharp edges, cracks, poor or unsound welds, or any other defect that could impair its performance or serviceability (see 4.5.1).

3.23.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to ensure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners except when sharp edges or corners are required or where they are not detrimental to safety.

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3.23.2 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or cap-screws shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.23.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the aerosol generator to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/4 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the aerosol generator shall be at the option of the contractor.

3.23.4 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Structural Welding Code, AWS D1.1, Qualification, Section 5.

Welding Qualifications of the ASME, Section IX.

Contractors who only make horizontal welds need not qualify welders for "all position welding." Subject to approval by the Government, the contractors standard welder qualification may be substituted in lieu of the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual nor does it commit the government to the acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection comparison (see 4.6).

d. Inspection of packaging (see 4.7).

4.3 Qualification inspection. The inspection will be performed by the Government at a site selected by the Government. Failure of the unit to pass the examination and tests shall be cause for exclusion from the Qualified Products List. The test unit may be a standard production unit from the contractor's current inventory provided the unit meets the requirements of the specification and is representative of the manufacturing technique applicable to the remaining units to be furnished under the contract.

4.3.1 Sample. One or more aerosol generators, as specified by the Government, shall be presented by the manufacturer for qualification inspection.

4.3.2 Examination. The aerosol generator shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.3 Tests. The aerosol generator shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. Acceptable quality levels (AQLs) shall be 4.0 percent defective for major defects and 6.5 percent defective for minor defects.

4.5 Inspection procedure.

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4.5.1 Examination. The aerosol generator shall be examined as specified herein for the following defects:

Major

101. Assembly not complete or not correct.
102. Parts or components missing or not as specified.
103. Material not as specified.
104. Lubrication devices not accessible or not as specified.
105. Electromagnetic interference suppression components and network not identical with approved model.
106. Safety provisions not as specified.
107. Engine HP rating not as specified.
108. Blower size rating not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification and stenciling marking not as specified.
203. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test preparation. Prior to the start of testing, the aerosol generator shall be prepared in accordance with the manufacturer's instructions.

4.5.2.2 Calibration. Two calibration tests shall be performed at a flow rate of 4.3 ounces per minute. One test will be conducted with chlorpyrifos, and the other test will be conducted with technical grade malathion. Tests shall be performed at 70 to 100°F ambient temperature. When specified (see 6.2), other flow rates shall be performed as directed. Flows shall be measured using a graduated cylinder and stopwatch. Each measured flow shall be within 10 percent of the value shown on the readout. A deviation of more than 10 percent shall constitute failure of this test. Failure of the manufacturer to provide for calibration in the design of the machine or failure to provide instructions for calibration shall also constitute failure of this test.

4.5.2.3 Road tests. Mount the aerosol generator on an appropriate vehicle and perform the following tests prior to the performance test specified in 4.5.2.4:

a. Unimproved roads. Drive the properly mounted aerosol generator 100 miles on unimproved roads at maximum safe speeds. The course shall consist of poorly maintained gravel and dirt roads with moderate washboarding, rutting, and occasional potholes. The aerosol generator will not be operated during this test.

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4.5.2.3 (cont'd)

b. Cross-country trails with aerosol generator operating. The properly mounted aerosol generator shall be started up and adjusted for routine operation. The aerosol generator shall be operated continuously (except for refueling) at designed operating engine speed for a 50-mile cross-country test in which the carrier vehicle will be driven at a speed of 10 miles per hour (mph). Technical grade malathion shall be dispersed through the nozzle for not less than 30 minutes during each hour of the road test. Inability of the aerosol generator to spray shall constitute failure of this test.

c. Failure of road tests. Examine the aerosol generator at the completion of each phase of the road test. The breaking or loosening of any fitting, seal, or other component that would cause the aerosol generator to malfunction or create a toxic hazard shall constitute failure of any specified road test.

4.5.2.4 Performance. The aerosol generator shall be started up and operated at designed engine speed and shall not exceed 80 percent of maximum rated horsepower. The machine shall be operated continuously for 8 hours except for refueling. The test formulation shall be technical grade malathion or mineral oil and shall be dispersed through the nozzle for at least 30 minutes during every 2 hours. Any malfunction of a component that stops or delays normal machine operation shall constitute a failure of this test.

4.5.2.5 Droplet size tests. Droplet size tests shall be conducted using chlorpyrifos and technical grade malathion as test insecticides. The engine of the aerosol generator shall be started up and operated at a power level not to exceed 80 percent of maximum rated horsepower. After 10 minutes of operation, atomization performance shall be determined by a Teflon® coated glass microscope slide-settlement chamber method. Droplet size shall be 7 to 17 microns volume median diameter (VMD). Aerosol generators that are outside this range by more than 10 percent with either test insecticide fail the droplet size test. Aerosol generators that are outside the 7 to 17 VMD range by not more than 10 percent shall be retested a second time. Aerosol generators that are outside the 7 to 17 VMD range with either test insecticide when retested fail the droplet size test. Procedures for the droplet size tests shall be as follows:

a. Use a 2 X 2 X 2 foot settlement chamber constructed of clean, unfinished plywood. The settlement chamber shall be airtight with a 2 X 2 foot door on one side for aerosol entry. The chamber shall be elevated so that the door opening is centered with the nozzles(s) of the aerosol generator.

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b. Place four Teflon® coated glass microscope slides on the floor of the settlement chamber for each aerosol collection. One slide shall be placed in the center of each of four quadrants of the floor area.

c. With the generator in operational mode, move in a straight line perpendicular past the settlement chamber at a speed of 5 to 10 mph so that the nozzle(s) comes within 3 to 6 feet of the door opening at the closest point.

d. Ambient temperature shall be 70 to 100°F, and natural wind currents shall not exceed 4 mph while moving the aerosol generator past the settlement chamber.

e. Close the door of the settlement chamber immediately after passage of the aerosol generator to prevent loss of the aerosol sample. Wait 15 minutes after closing the door and then remove glass slides.

f. Flow rate for the test insecticides shall be 4.3 fluid ounces per minute for technical grade malathion and chlorpyrifos.

g. Measure the diameter of 100 droplets per slide with minimum diameter measurement being the ocular division nearest 4 microns. Volume median diameter will be determined from the aggregate number of droplets collected from the four coated slides for each insecticide.

h. Correct measured diameters for spread on Teflon® and then calculate VMD on the basis of volume (diameter cubed).

i. Repeat collection procedure one time for each test insecticide. Droplet size criteria and test procedures shall be the same as the first droplet size test for each insecticide.

4.5.2.6 Electromagnetic interference when specified. The unit shall be tested to determine compliance with 3.13.1. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.2.7 Noise level measurement.

4.5.2.7.1 Personnel noise testing. The aerosol generator shall be tested in accordance with the provisions of MIL-STD-1474 for steady-state noise, personnel-occupied areas. Noise levels shall be taken while the system is at maximum rated output. The data should include dB(A) levels taken at all locations normally occupied by personnel during start-up and operation. Duty cycle testing shall not be used.

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4.5.2.7.2 Exterior noise testing. The aerosol generator shall be tested in accordance with the provisions for exterior noise of MIL-STD-1474. Test procedure SAE J1174 may be used as an alternate procedure.

4.5.2.7.3 Hazard precautions. Should the noise level exceed 85 dB(A) at any operator position, and the excess noise level can be documented as being clearly beyond the state-of-the-art, failure to provide the precautions specified in 3.6.2 shall constitute failure of this test.

4.5.2.7.4 Sequence of noise level measurement testing. When reliability tests (see 4.5.2.8) are performed, the noise level measurement shall be executed at their conclusion.

4.5.2.8 Reliability. Type I aerosol generators shall be tested as specified in 4.5.2.4 for 400 hours, and type II aerosol generators shall be tested as specified in 4.5.2.4 for 200 hours. If three or fewer failures occur, the unit shall be considered acceptable. In addition to the overall limit of three failures, each test machine (in accordance with the definitions that follow) shall not experience: (a) any catastrophic failures, (b) more than one major failure, (c) more than three minor failures. Note that a major failure cascades into the minor failure category; that is, a major failure would reduce the allowed number of minor failures by one. The Government may require the manufacturer to make minor modifications to correct the cause of any failure identified in the reliability testing, or the manufacturer may volunteer modifications to prevent failure reoccurrence during testing. Any such modifications are subject to Government approval and shall be accomplished at no cost to the Government. After completion of such modifications, the Government shall determine if any tests need to be repeated or extended. After successful corrective action has been completed, at the discretion of the Government, failures may be eliminated from the count if there is convincing evidence that the cause of the failure has been eliminated. The following definitions apply to reliability failure criteria:

a. General definition for failure - Any malfunction that causes inability to commence operation, cessation of operation, degradation below the operating conditions required to continue the test, or a safety hazard to personnel.

b. Catastrophic failure - A serious failure that totally disables the machine and cannot be repaired in less than 8 man-hours or requires part replacements that exceed 25 percent of the cost of the machine.

c. Major failure - A failure that requires 1 to 8 man-hours to repair or requires part replacements that cost between 3 percent and 25 percent of the machine cost. Sophisticated repair equipment is permitted in this category.

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4.5.2.8 (cont'd)

d. Minor failure - A failure that can be corrected in less than 1 man-hour without the use of sophisticated equipment and requires part replacements that do not exceed 3 percent of machine cost.

e. Sophisticated repair equipment - Machine shop type support such as welding, cylinder boring, lathe work, special riveting, etc.

f. Machine cost - The Government's cost per unit.

g. Parts costs - The Government's cost at time of failure.

4.5.2.9 Maintenance support evaluation. Maintenance operations required during testing shall be accomplished to determine conformance to 3.12. Failure of the support elements to maintain the aerosol generator as specified herein shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the mean time between failures (MTBF).

4.6 Inspection comparison. The Government may select aerosol generators at any time during the contract period and subject these aerosol generators to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on aerosol generators selected at random from those that have been accepted by the Government and will not include the previously qualified aerosol generators. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison; and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison aerosol generator to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison aerosol generator and shall be cause for refusal by the Government to continue acceptance of production aerosol generators until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on aerosol generators previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered *prima facie* evidence that all aerosol generators accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.7 Inspection of preparation for delivery. The preservation, packaging, packing, marking, and quality conformance of each aerosol generator shall be inspected in accordance with the requirements of MIL-S-12651. Unless otherwise specified (see 6.2), a preproduction pack shall be furnished within the time frame required for examination and testing in accordance with MIL-S-12651.

5. PACKAGING

5.1 Preservation, packing, and marking. The preservation, packing, and marking shall be in conformance with MIL-S-12651. The level of preservation and the level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The aerosol generator is intended primarily for mosquito control through dispersal of concentrated insecticides at a ULV rate from truck or jeep.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When gauges will be provided for insecticide and fuel tanks (see 3.2).

c. When sample units will be required for qualification inspection and number of units (see 4.1.1 and 4.3.1).

d. When qualification inspection will be accomplished by the Government.

e. When electromagnetic interference suppression is required (see 3.13.1).

f. When elapsed time indicator will be required (see 3.17.1).

g. Color required (see 3.20).

h. When welding process will not be at the option of the contractor (see 3.23.3).

i. When and which additional flow rates will be required during calibration (see 4.5.2.2).

j. Time frame required for submittal of preproduction pack or when a preproduction pack is not required (see 4.7).

k. Level of preservation and level of packing required (see 5.1).

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6.3 Consideration of data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data Requirements	Applicable DID No.	Opts.
6.2.2.1	Commercial Manuals	DI-M-4022C	

(Data Item Descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3.1 Data. The contracting officer should make provisions in the contract that the manufacturer provide a detailed operator's manual with each unit which includes installation, maintenance and operational instructions, and safety precautions (see 3.17.1.1). The contracting officer should also assure that the manufacturer provides a detailed parts list with each unit, applicable to the model furnished, which includes parts numbers, the supply source for each, and schematic of electrical and electronic circuitry.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-52940 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered in this specification. The activity responsible for the Qualified Products List is the U.S. Army Biomedical Research and Development Laboratory, ATTN: SGRD-UBE-D, Fort Detrick, Frederick, MD 21701-5010; and information pertaining to qualification of products may be obtained from that activity.

6.5 Noise limits. When the noise level of the aerosol generator exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS paragraph) must include the requirement for hearing protection, the noise level of the aerosol generator, the distance from the aerosol generator at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

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6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions on shipment of the aerosol generator.

6.7 Disposability. One or more of the following methods shall be used to accomplish disposal of the packaging material: reuse, recycling, sanitary landfill, incineration, or pyrolysis.

6.8 Recycled material. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new, and shall be fabricated using materials produced from recycled materials to the maximum extent practicable without jeopardizing the intended use. The term "recycled materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically authorized (see 3.4).

6.9 Subject term (key word) listing.

Aerosol generator
Calibration
Droplet sizes
Examination
Flow rate
Insecticide
Maintenance ratio
Major failure
Mean time between failures
Microns
Military specification
Minor failure
Qualified Products List
Reliability
Tests
Ultra low volume

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - MD
Navy - YD
Air Force - 99

Review activities:
Army - CE
Navy - MC, MS
Air Force - 84
DLA - CS
AFPMB - PM

Preparing activity:
Army - MD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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